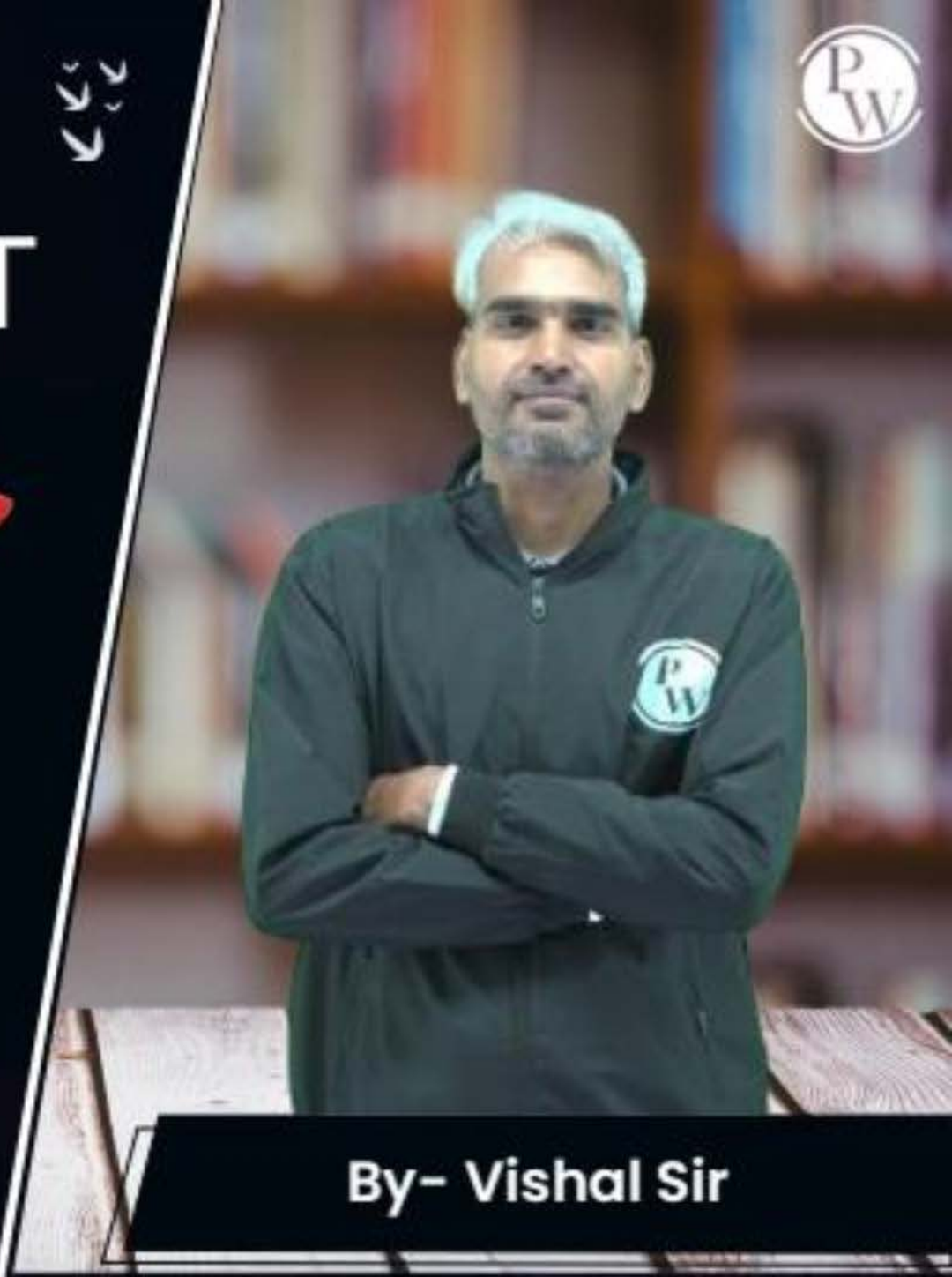
Computer Science & IT



Database Management System

Query Languages

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture



- Topic: Decomposition of relation
- Topic: Multivalued dependency
- Topic: Properties of multivalued dependency
- Topic: Fourth normal form (4NF)

Slide

Topics to be Covered

Query languages
CH-2



- Topic — Query Languages
- Topic — Relational algebra
- Topic — Basic relational algebra operations
- Topic — Number of super key concept

CH-1

Slide



Topic : Query languages

Query languages

Procedural query languages

We define that what to retrieve from the database and we also define the procedure to retrieve the desired data from database

eg: Relational Algebra

Non-procedural query languages

We define, what to retrieve from the database, and we use syntax provided by the languages to access the desired data from the database

eg: Structured query language (SQL)
Tuple relational Calculus (TRC)
Domain relational Calculus (DRC)

Note:- Query Condition evaluates tuple by tuple, Only one tuple at a time.

∴ If we want to compare two or more tuples of the same table or different tables then, we need to Combine those tuples into a single tuple so that we can evaluate the query condition.

{ To combine tuples into a single tuple we can use Cross product.



Topic : Relational Algebra

Relational Algebra is a procedural query language used to query the relational database tables to access data.

Relational Algebra operation can be classified into two types:

1) Basic Relational Algebra Operations ✓
2) Derived Relational Algebra operations ✓

↳ i.e., they are derived using one or more basic relational algebra operations



Topic : Basic Relational Algebra operators

1. Projection(π)
2. Selection (σ)
3. Cross Product ($\times$)
4. Union (U)
5. Set Difference (−)
6. Rename (ρ)

← Set Operations



Topic : Derived Relational Algebra operators

1. Intersection (∩) ← Set operations
2. Join Operations ("⋈")
3. Division Operation (÷)



Topic : Projection (π)

w.r.t specified attributes

It is used to project the column data from a relation based on the attributes specified with projection operation.

e.g., $\pi_{<\text{attribute list}>}(R)$

Projection operator does not obey commutative property i.e.

$$\pi_{<\text{list2}>}(\pi_{<\text{list1}>}(R)) \neq \pi_{<\text{list1}>}(\pi_{<\text{list2}>}(R))$$

$$\pi_{Sid}(\pi_{Sid,Cid}(R)) \neq \pi_{Sid,Cid}(\pi_{Sid}(R))$$

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_1	IT
S_3	C_2	CS

① $\pi_{Sid, Cid, Branch}(R) \equiv (R) \equiv R$

The output will Contain all attributes and all tuples of relation R

② $\pi_{Sid}(R) =$

Sid
S_1
S_2
S_3

③ $\pi_{Cid}(R) =$

Cid
C_1
~~C_1~~
C_2

Relational algebra Operations will always Produce distinct tuples

Output of SQL query may contain duplicate tuples.

④ $\Pi_{Cid, Branch}(R)$ =

Cid	Branch
C_1	CS
C_1	IT
C_2	CS

tuple wise they are distinct



Topic : Selection(σ)

It is used to select the tuples from underlying relation based on the predicate condition specified with selection operation.

$$\sigma_{<selection_condition>}(R)$$

Selection operator is commutative in nature i.e.

w.r.t "AND"

$$\sigma_{A \wedge B}(R) = \sigma_{B \wedge A}(R)$$

OR

$$\sigma_{B}(\sigma_{A}(R)) = \sigma_{A}(\sigma_{B}(R))$$

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_1	IT
S_3	C_2	CS

* Select the tuple with $Cid = C_1$

$$\sigma_{(Cid = C_1)}(R)$$

=

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_1	IT

* Retrieve Sids of the students with $Cid = C_1$

$$\pi_{Sid}\left(\sigma_{Cid = C_1}(R)\right)$$

③ ② ①

Order of execution

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_1	IT
S_3	C_2	CS

Retrieve Sid of the student with Cid = C_1 and Branch = CS

① $\pi_{Sid}\left(\sigma_{(Cid = C_1 \wedge Branch = CS)}(R)\right)$

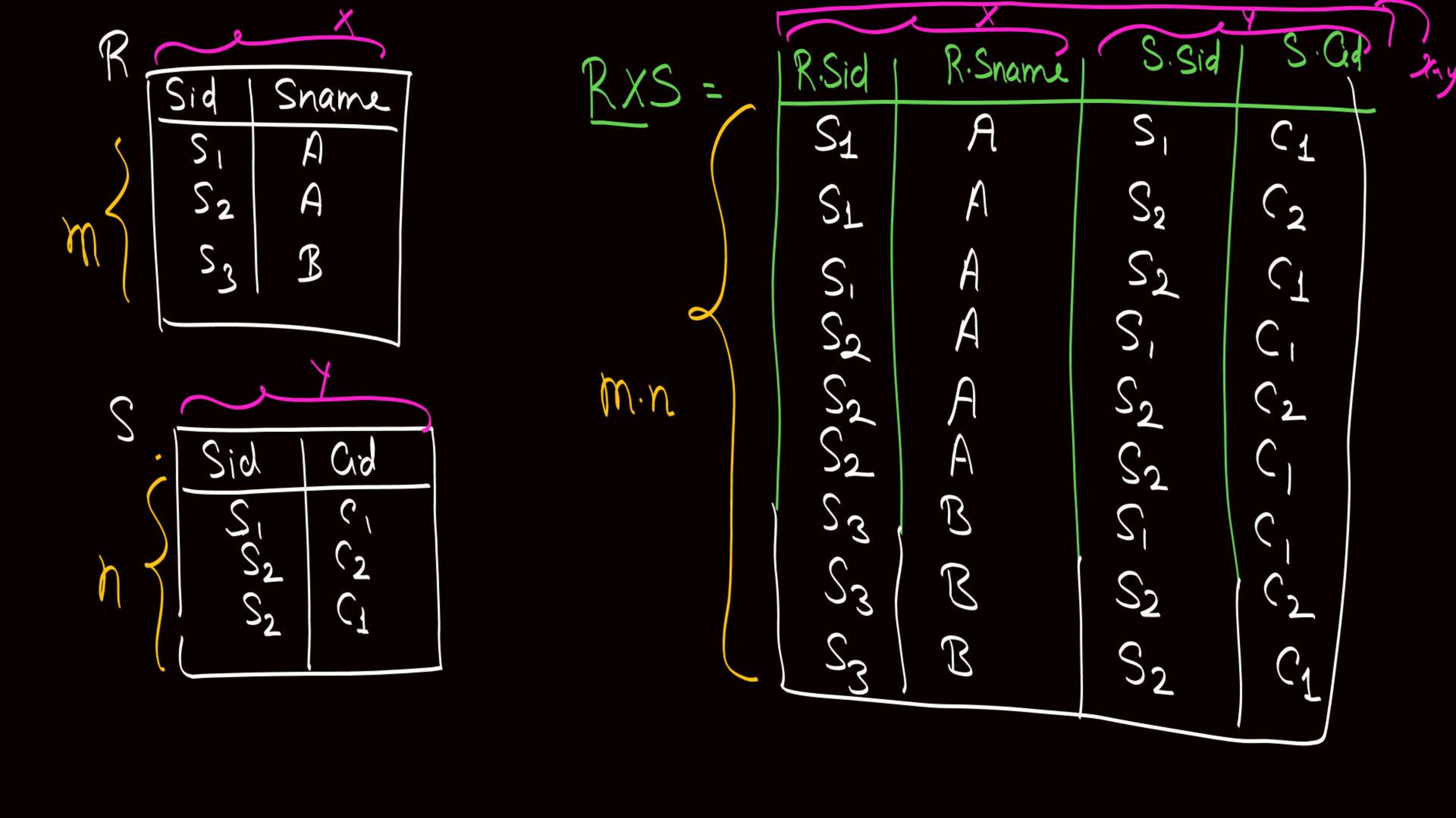
② $\pi_{Sid}\left(\sigma_{Branch = CS}\left(\sigma_{Cid = C_1}(R)\right)\right)$

③ $\pi_{Sid}\left(\sigma_{Cid = C_1}\left(\sigma_{Branch = CS} R\right)\right)$



Topic : Cross Product (×)

Cross-product is a binary operation. Let R and S are any two relation, then cross product R × S will result in all attributes of R followed by all attribute of S with all possible combinations of tuples from R and S.


R
X
Sid
Sname
S1
A
S2
A
S3
B
m
S
Y
Sid
Cid
S1
C1
S2
C2
S2
C1
n
R X S =
m·n
R.Sid
R.Sname
S.Sid
S.Cid
X+Y



Topic : Union, Set difference, Intersection

- Union, Set Difference and Intersection are the Set operations.
- To use set theory operators on any two relations, those relations must be union compatible.
- The union compatibility of relations implies that the participating relations must fulfil the following conditions.

1. Same degree, i.e. The two relations must have the same number of attributes.
2. Same domain of each corresponding attributes of relations

let R =

Sid	Sname

S

Sid	Sname	Branch

No. of attributes in R ≠ No. of attributes in S

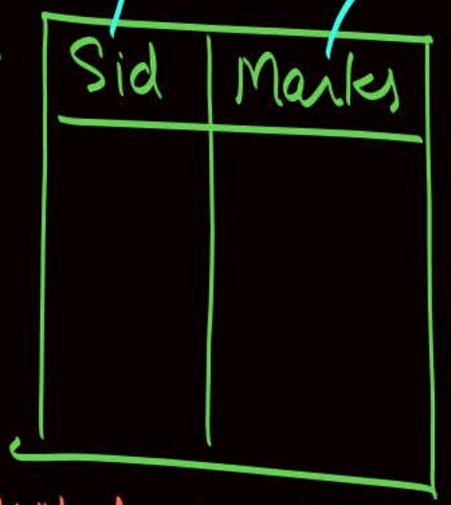
∴ R & S are no Union Compatible

↳ i.e. Union, Set difference & Intersection can not be applied on R & S.

Let R =



S =



✓ ① No. of attributes in R = No. of attributes in S.

✗ ② Domain of the 2nd attribute of R and S does not match

∴ Not Union Compatible



① No. of attributes in R = No. of attributes in S.

② Domain of the corresponding attributes of R & S is same

∴ Both are valid, i.e. R & S are Union Compatible

* If we perform the set operations on R and S then the resulting relation will take the names of its attributes from the relation on L.H.S. of set op^n {i.e. if R ∪ S then names of attributes will be same as attributes of R}



Topic : Union, Set difference, Intersection

- Union (A ∪ B)- It contains all tuples from each of the relations.
{but no duplication}

- Difference (A – B)- It contains all the tuples that are contained in the relation A but are not present in the relation B

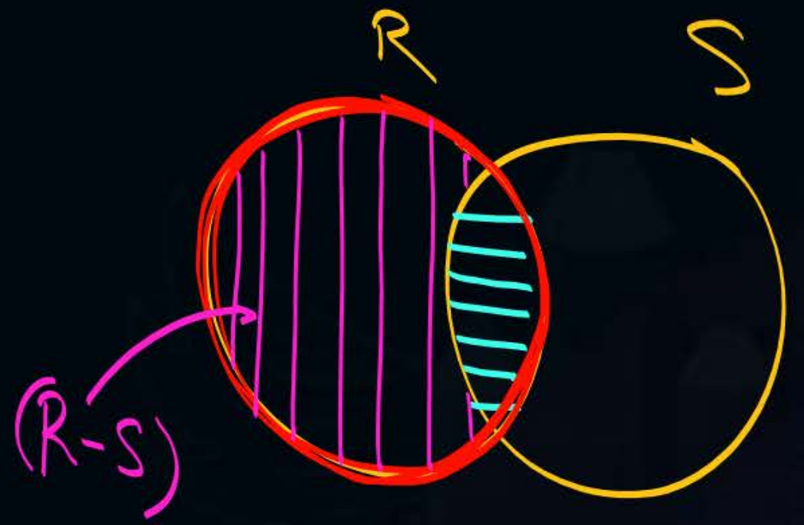
- Intersection (A ∩ B)- It contains all the tuples that are contained in both the relations A and B.



Topic : Intersection

* Intersection is a derived relational Algebra Oph. which is derived using set difference

$$R \cap S = R - (R - S)$$

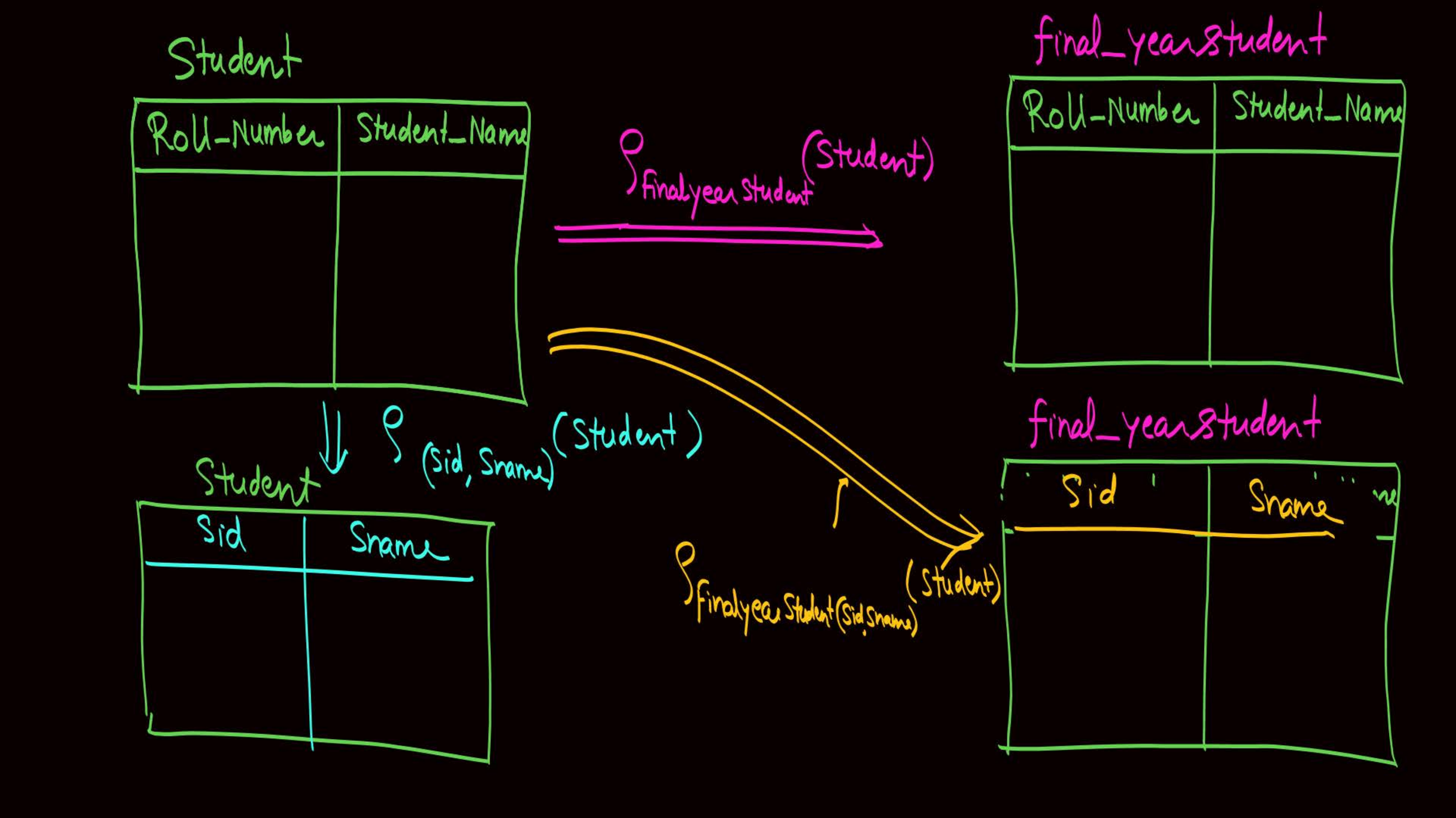




Topic : Rename (ρ)

Rename operation can be used to rename attributes of the relation, name of the relation or both.

- Renaming a relation: $\rho_{FinalYrStudents}(Students)$
- Renaming attributes: $\rho_{(SID,\ Sname)}(Students)$
- Renaming both: $\rho_{FinalYrStudents\ (SID,\ Sname)}(Students)$


Student
Roll_Number
Student_Name
$\rho_{finalyear\ Student}(Student)$
final_yearstudent
Roll_Number
Student_Name
$\rho_{(Sid, Sname)}(Student)$
Student
Sid
Sname
$\rho_{finalyear\ Student(Sid, Sname)}(Student)$
final_yearstudent
Sid
Sname



Topic : Number of super keys

Let $R(A_1, A_2, A_3, \ldots A_N)$ be the relational schema,

How many superkeys are possible for relation R if,

① (A_1) is the only candidate key of Relⁿ R

- Superkey = All attributes of at least one of the C.K + 0 or more attributes from remaining attributes
- Any Superset of C.K is a Super key



Topic : Number of super keys

Let $R(A_1, A_2, A_3, \ldots A_N)$ be the relational schema,

How many superkeys are possible for relation R if,

(1) (A_1) is the only candidate key of Relⁿ R

$R(A_1, A_2, A_3, A_4, A_5, \ldots A_N)$

Choices wrt Super key of R $= 1 * \underbrace{2 * 2 * 2 * 2 \ldots * 2}_{(N-1) \text{ times '2'}} = 2^{N-1}$

A_1 Must be Present



Topic : Number of super keys

Let $R(A_1, A_2, A_3, \ldots A_N)$ be the relational schema,

How many superkeys are possible for relation R if,

(2) (A_1A_2) together is the only candidate key of Relⁿ R

Superkey = (A_1+A_2) + Any subset of remaining attributes

Remaining attributes are = N-2

∴ 2^{N-2} different subsets are possible

∴ Total No. of superkeys = 2^{N-2}



Topic : Number of super keys

Let $R(A_1, A_2, A_3, \ldots A_N)$ be the relational schema,

How many superkeys are possible for relation R if,

(2) (A_1A_2) together is the only candidate key of Relⁿ R

$R(A_1, A_2, A_3, A_4, A_5, \ldots A_N)$

Choices wrt Super key of R = $1 \times 1 \times \underbrace{2 \times 2 \times 2 \ldots \times 2}_{(n-2) \text{ times } '2'} = 2^{N-2}$

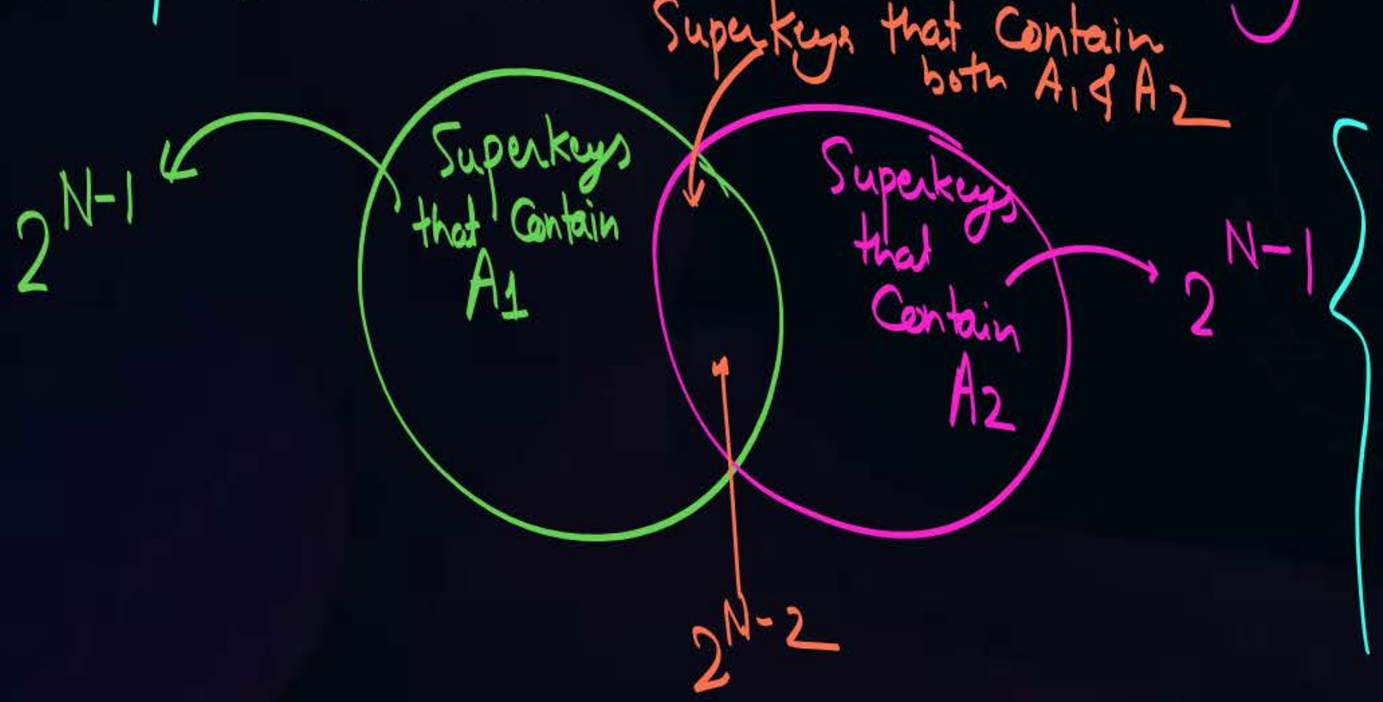
Must be Present



Topic : Number of super keys

Let $R(A_1, A_2, A_3, \ldots A_N)$ be the relational schema, How many superkeys are possible for relation R if,

(3) A_1 & A_2 are the candidate keys of Relⁿ R



∴ Total no. of superkeys

$= 2^{N-1} + 2^{N-1} - 2^{N-2}$

$= 2 \cdot 2^{N-2} + 2 \cdot 2^{N-2} - 2^{N-2}$

$= 3 \cdot 2^{N-2}$

Soln ②

$R(A_1\ A_2\ A_3\ A_4\ A_5 \ldots\ldots A_N)$

$1 * 1 * 2 * 2 * 2 * \ldots\ldots * 2 = 2^{N-2}$

Choices w.r.t S.K =

$1 * 1 * 2 * 2 * 2 * \ldots\ldots 2 = 2^{N-2}$

or $1 * 1 * 2 * 2 * 2 * \ldots\ldots 2 = 2^{N-2}$

or $1 * 1 * 2 * 2 * 2 * \ldots\ldots 2 = 2^{N-2}$

Total No. of Superkey $= 3 \cdot 2^{N-2}$

Soln ③:

- Total no. of subsets of attributes of $R = 2^N$

- Number of subsets in which neither A_1 nor A_2 is present $= 2^{N-2}$

Total subsets − Subsets in which neither A_1 is present nor A_2 is present

Subsets in which either A_1 or A_2 or both A_1 & A_2 are present {i.e Super key}

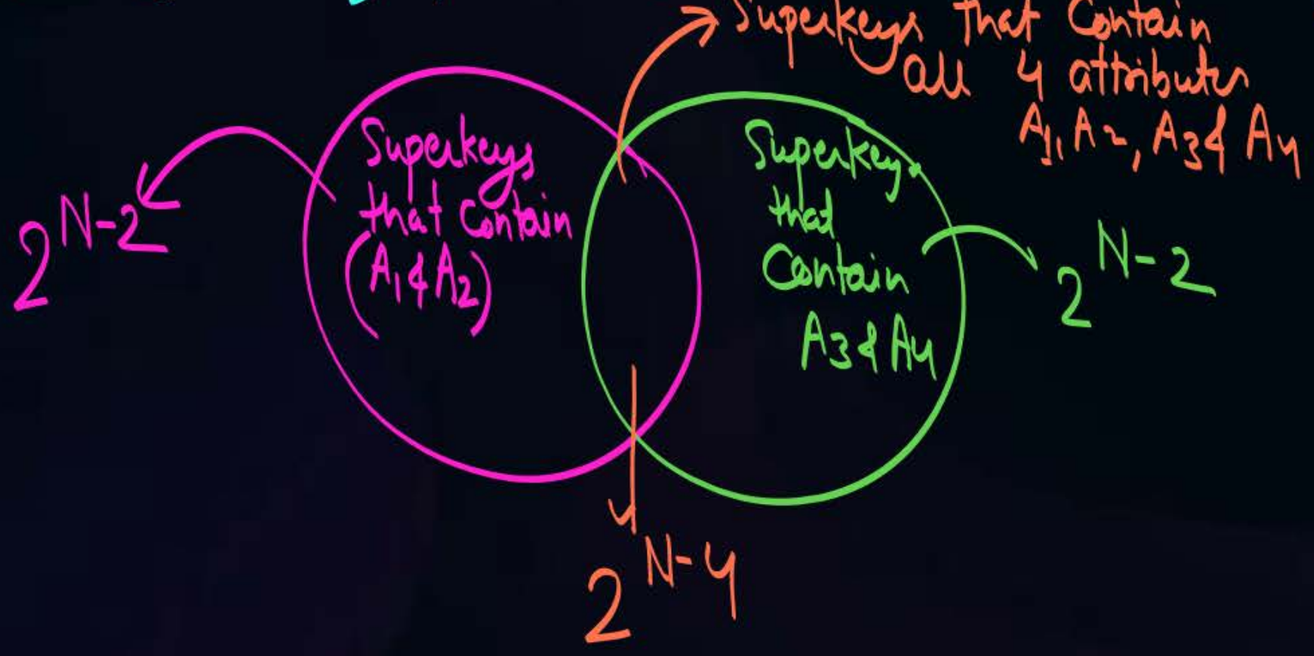
$= 2^N - 2^{N-2} = 4.2^{N-2} - 2^{N-2} = 3.2^{N-2}$

Topic : Number of super keys

Let $R(A_1, A_2, A_3, \ldots A_N)$ be the relational schema,

How many superkeys are possible for relation R if,

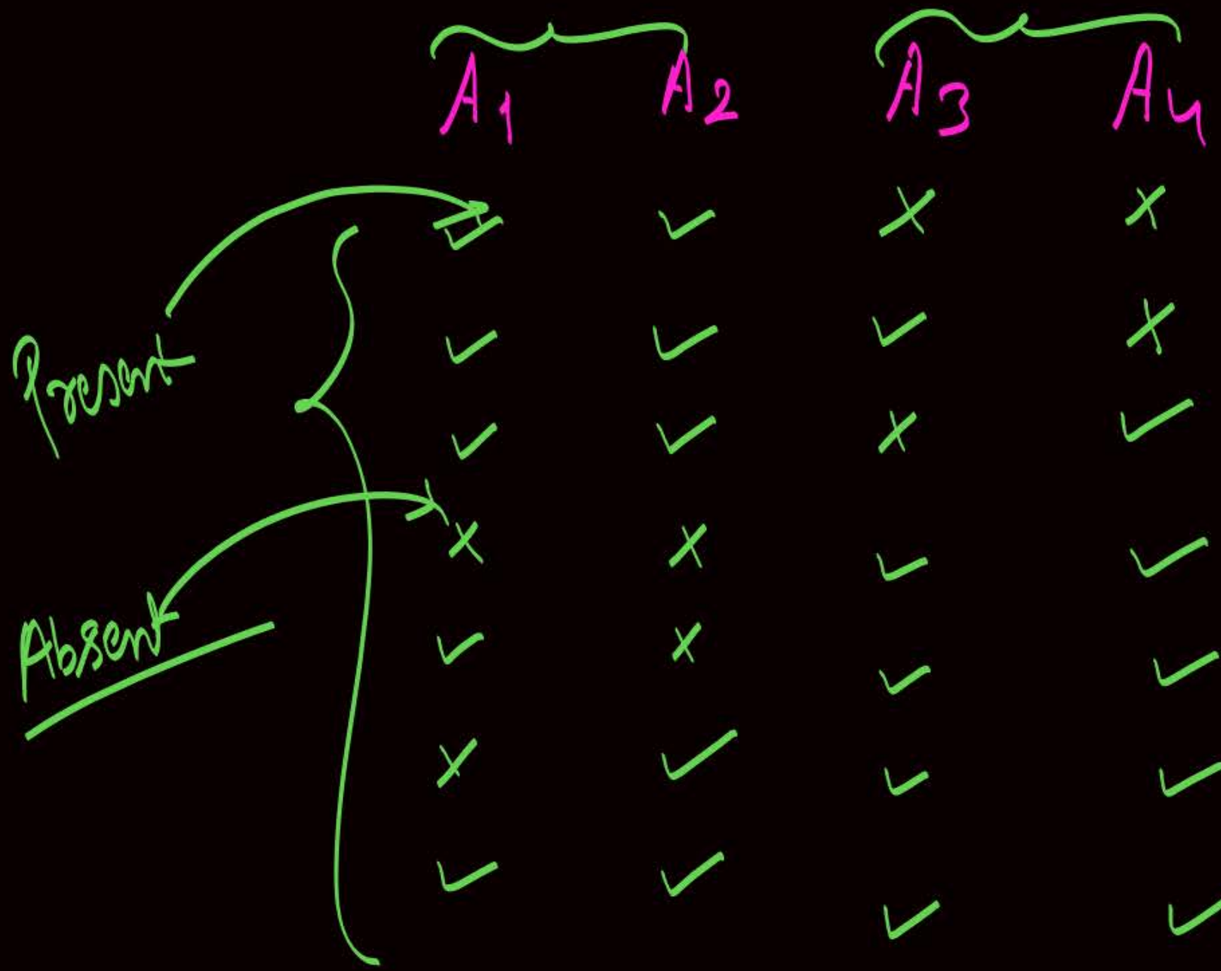
(4) $(A_1\ A_2)$ & $(A_3 A_4)$ are the candidate keys of Relⁿ R



Total No. of superkeys

$= 2^{N-2} + 2^{N-2} - 2^{N-4}$

$= 7 \cdot 2^{N-4}$



$A_5 \; - \; - \; - \; A_N$

$2 \times \cdots \cdots 2 = 2^{N-4}$

$2 \times 2 \times \cdots 2 = 2^{N-4}$

$2 \times 2 \times \cdots 2 = 2^{N-4}$

$= 2^{N-4}$

$= 2^{N-4}$

$= 2^{N-4}$

$= 2^{N-4}$

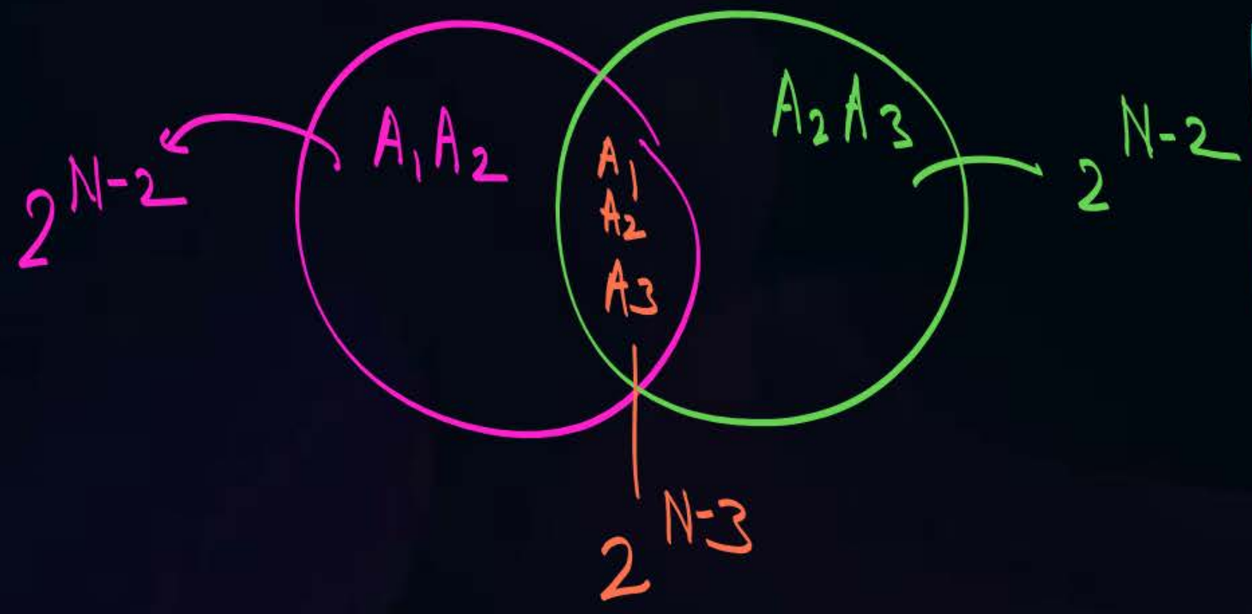
$= 2^{N-2}$

$7 \cdot 2^{N-4}$

Topic : Number of super keys

Let $R(A_1, A_2, A_3, \ldots A_N)$ be the relational schema, How many superkeys are possible for relation R if,

(5) $(A_1\ A_2)$ & $(A_2 A_3)$ are the candidate keys of Relⁿ R



Total No. of Superkeys

$= 2^{N-2} + 2^{N-2} - 2^{N-3}$

$= 3 \cdot 2^{N-3}$

A_1 A_2 A_3 A_4 - - - - A_N

✓ ✓ x 2 x - - - 2 = 2^{N-3}

x ✓ ✓ 2 x - - - 2 = 2^{N-3}

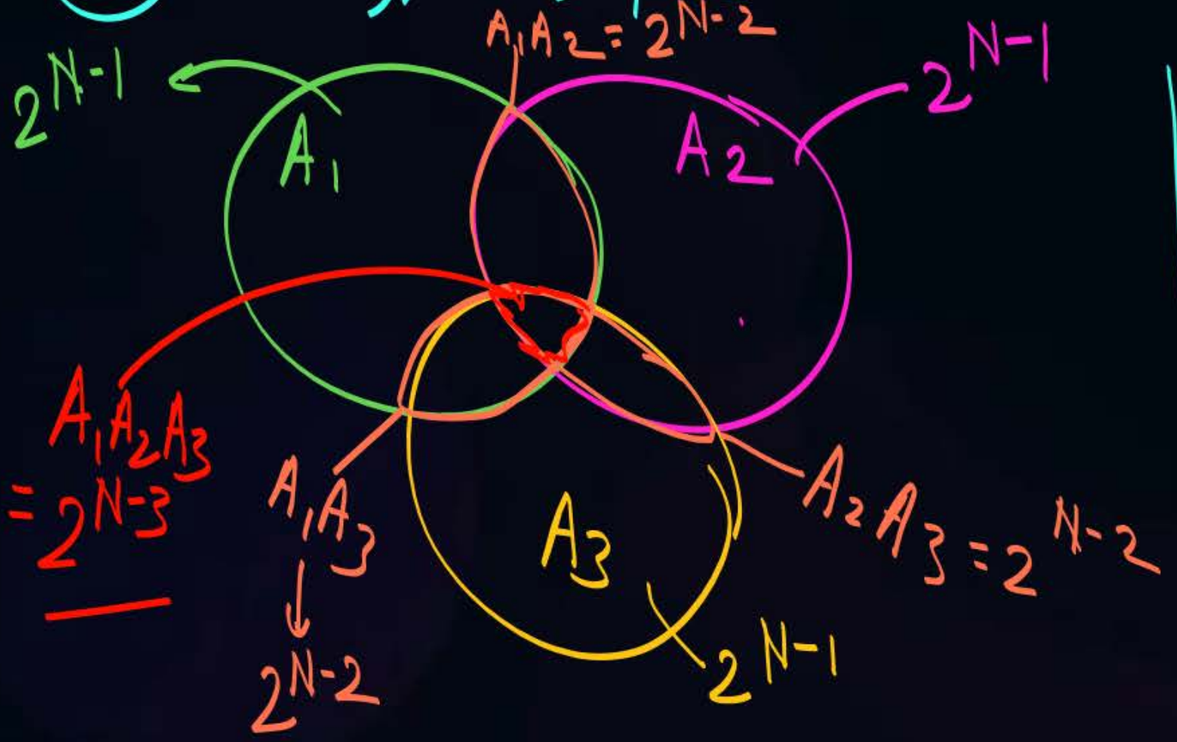
✓ ✓ ✓ 2 x - - - 2 = $\frac{2^{N-3}}{3 \cdot 2^{N-3}}$

Topic : Number of super keys

Let $R(A_1, A_2, A_3, \ldots A_N)$ be the relational schema,

How many superkeys are possible for relation R if,

(6) (A_1), (A_2) & A_3 are the candidate keys of Relⁿ R



Total No. of Super keys :-

$$2^{N-1} + 2^{N-1} + 2^{N-1} - 2^{N-2} - 2^{N-2} - 2^{N-2} + 2^{N-3}$$

$$= 7 \cdot 2^{N-3}$$



Topic : Number of super keys

Let $R(A_1, A_2, A_3, \ldots A_N)$ be the relational schema,

How many superkeys are possible for relation R if,

all attributes of R are Candidate keys

If all attributes are Candidate keys, then any subset of attributes will be a superkey

$\therefore 2^N - 1$ superkeys are possible.

2^N — Total no. of subsets

-1 — Except ϕ



2 mins Summary

- **Topic** Query Languages
- **Topic** Relational algebra
- **Topic** Basic relational algebra operations
- **Topic** Number of super key concept

THANK - YOU